

Project Overview

Developed a full-stack message storage application using HTML, CSS, JavaScript, Python Flask, and MySQL. The project follows a 3-tier architecture with clear separation between frontend, backend, and database layers.

Development (using AI)

Built an interactive frontend using HTML, CSS to handle user interactions and UI design. Implemented backend logic using Python Flask to manage APIs, requests, and business logic.

Used MySQL as the database to store and manage application data efficiently.

Version Control

Managed the entire source code using Git and GitHub.

Tracked changes, maintained repository history, and enabled collaborative development practices.

CI/CD Automation

Integrated Jenkins to build a CI/CD pipeline for automated workflow.

Automated code build, testing, and deployment whenever changes were pushed to GitHub.

Containerization

Used Docker and Docker Compose to containerize the application.

Managed frontend, backend, and database as separate services for consistent deployment across environments.

DevOps Implementation

Automated the complete deployment process to reduce manual effort and improve efficiency.

Ensured faster delivery and reliable application deployment using DevOps best practices.

Conclusion

This project helped in understanding real-world full-stack development along with DevOps tools and automation. It improved practical knowledge of CI/CD pipelines, containerization, and production-level deployment workflows.

A 3-Tier Full Stack Application with frontend, backend, and database architecture integrated using **DevOps CI/CD automation**.

Used technologies:

- 1.Github**
- 2.Jenkins(CICD)**
- 3.Docker**
- 4.Python-flask(Backend)**
- 5.HTML and CSS(Frontend)**
- 6.Mysql(Database)**

First, clone the project repository from <https://github.com/chandusaigari/project> and open it in VS Code.

After making changes, push the code to your GitHub repository using the following

Git commands:

```
git init
git clone https://github.com/chandusaigari/project.git
git branch -M main
git status
git remote add origin https://github.com/your_username/project
git add .
git commit -m "code committed"
git push origin main
```

Now the code will be available in your GitHub repository.

If your repository already contains files, use `git pull origin main` before pushing the code, because the files in github may not contain in your local vscode folder.

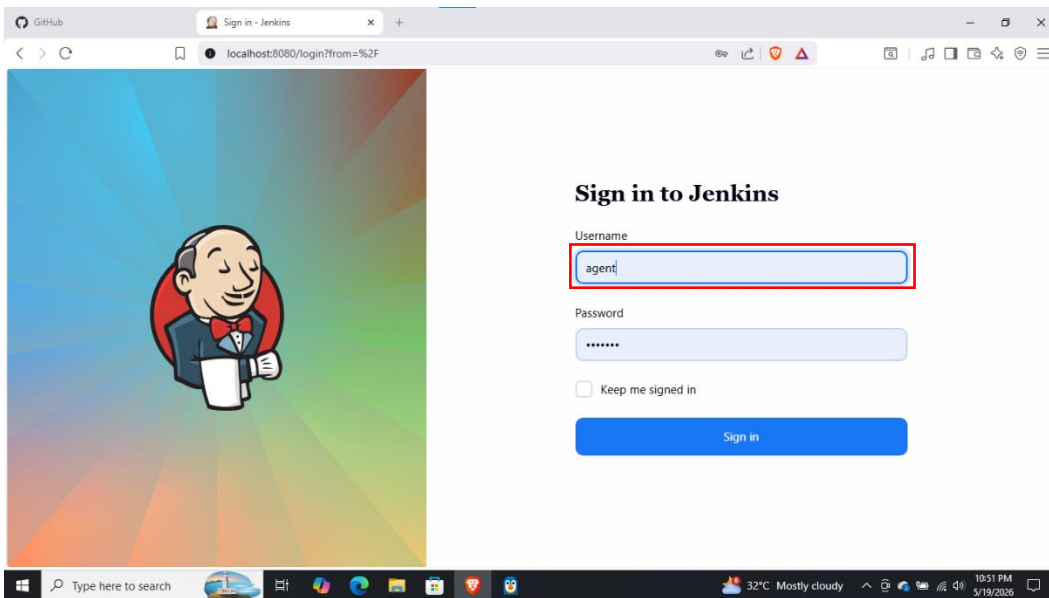
Make sure you downloaded Jenkins and docker for this project without those , this project would not happen .

what to do next after pushing to github?

After installing successfully Jenkins and docker do these below steps,
Open any distro of linux like Ubuntu latest version.

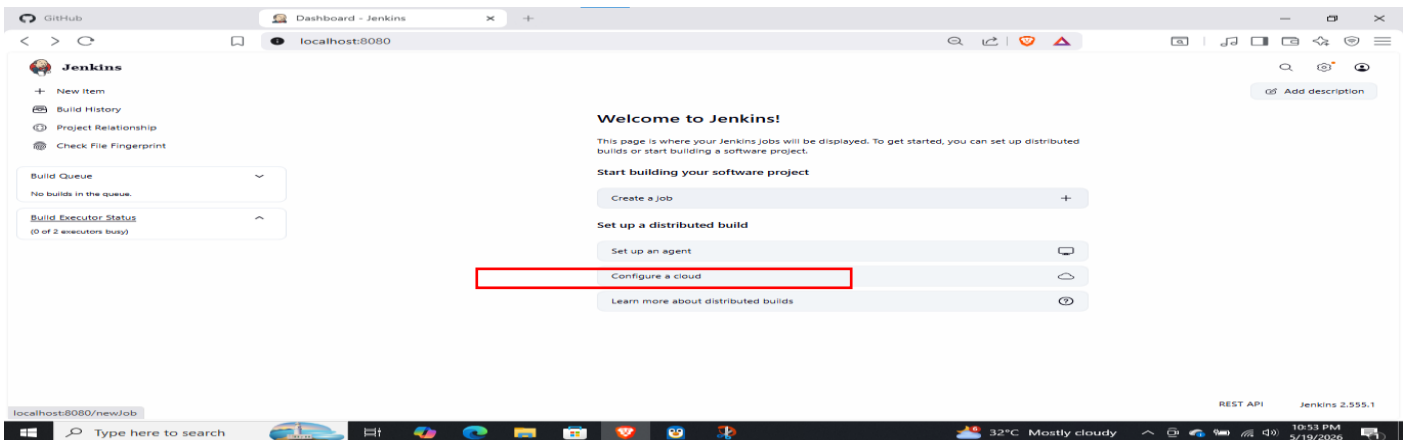
Type the command for starting Jenkins as shown in below picture

After this your Jenkins will start and you can access Jenkins from browser with
localhost://8080.



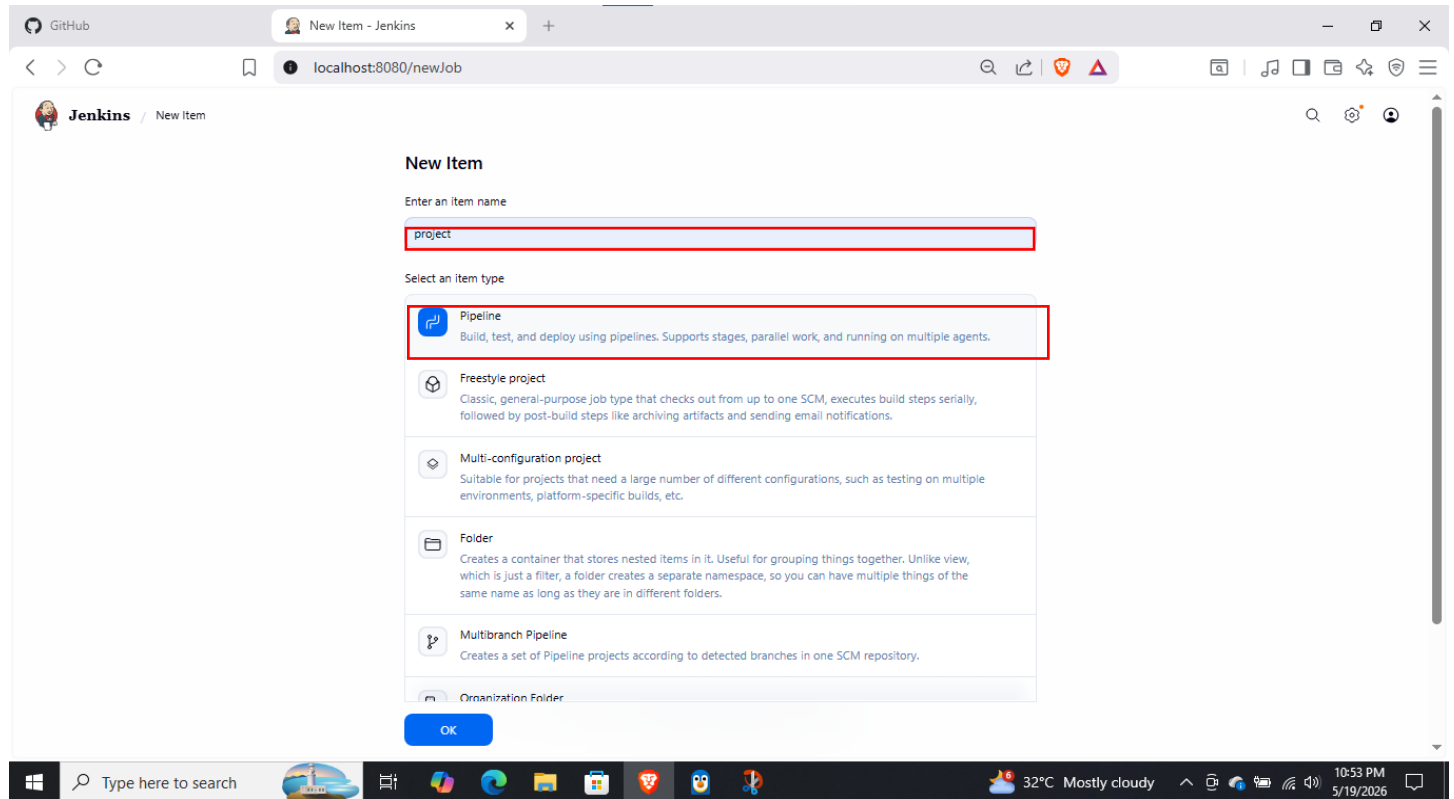
You need to sign in with your Jenkins **username** and **password**

After sign in you will get the Jenkins dashboard like this below one.



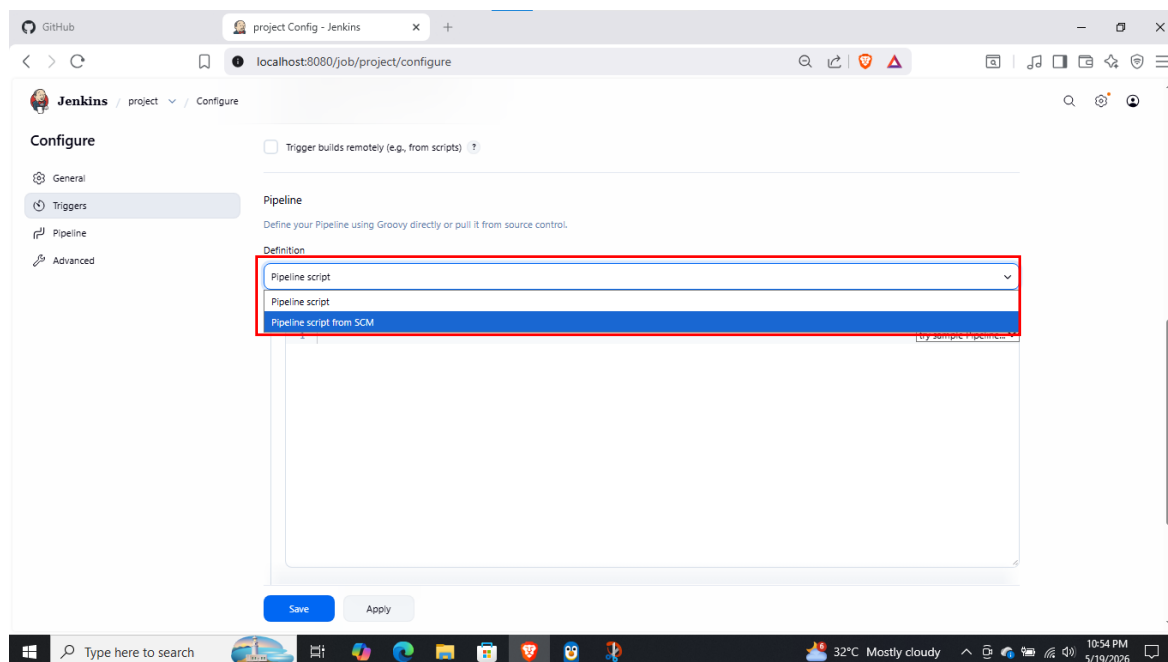
Build your First Job by clicking the above mentioned button

The Outcome will be :

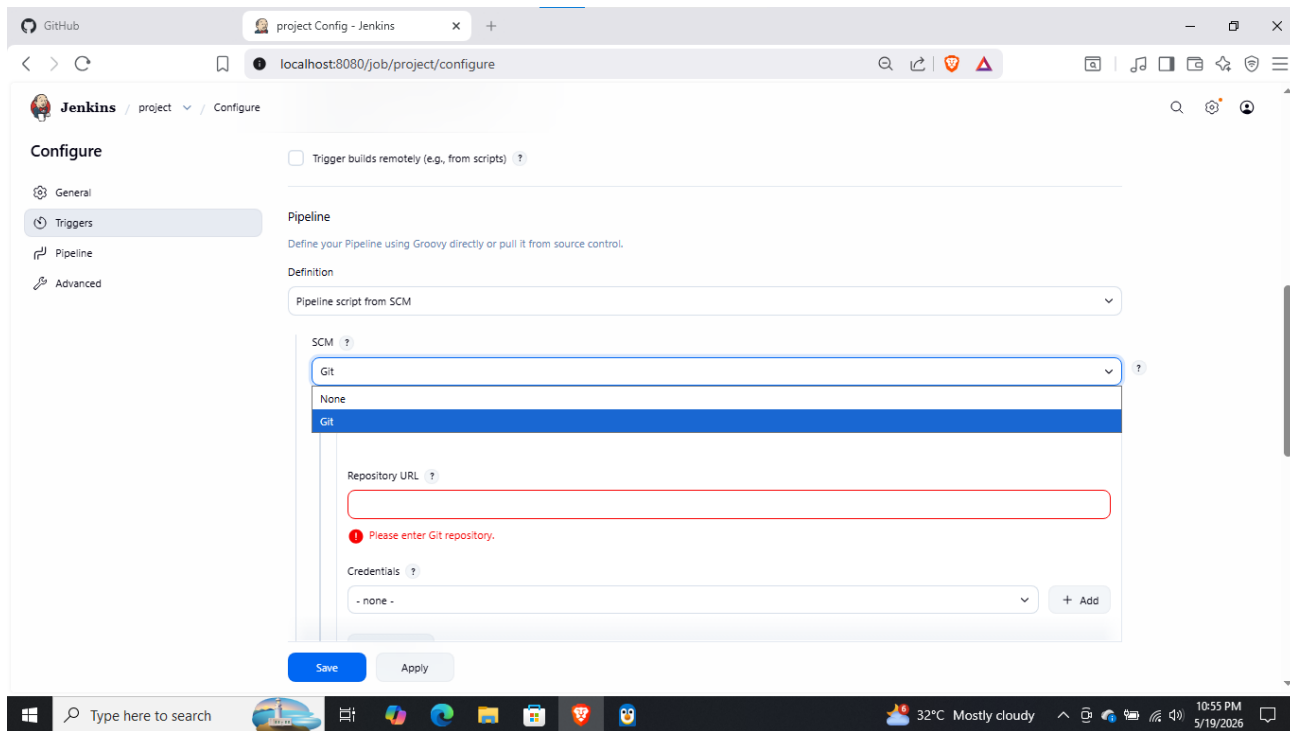


Make sure that **your repository name & Job name should be same** , select the pipeline

After saving ,you will be redirect to configure page then you need to choose the **pipeline script from scm** .

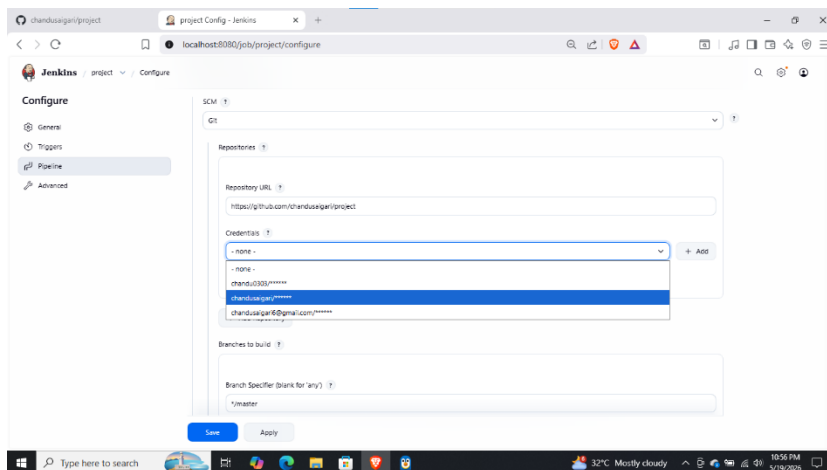


Now select the scm- source code management as Git



Now add the URL of your project in the Repository URL

If your project repo is not private no need to add credentials if it is private make sure you need to add github credentials.



Select the branch of your repo based on your project either it is main or master branch.

After applying and save , you can build the Job by pressing the button

BUILD NOW

The screenshot shows the Jenkins web interface for a job named 'project'. The left sidebar contains a menu with options: Status, Changes, Build Now (highlighted), Configure, Delete Pipeline, Full Stage View, Stages, Rename, and Pipeline Syntax. The main content area shows the 'Stage View' with a message: 'No data available. This Pipeline has not yet run.' Below this is a 'Permalinks' section. At the bottom left, there is a 'Builds' section showing 'No builds'. The bottom of the browser window shows the Windows taskbar with the search bar and various application icons. The system tray shows the date and time as 10:58 PM on 5/19/2026.

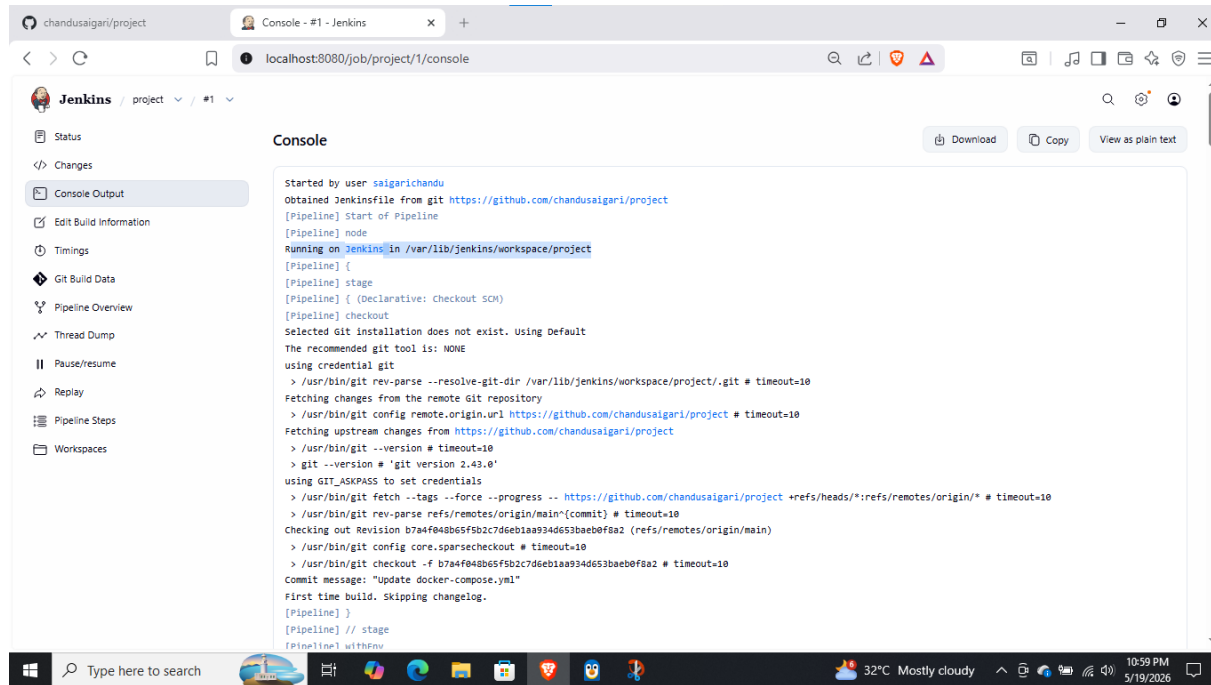
As you can see, by clicking **BUILD NOW** your job will get started..

The screenshot shows the Jenkins web interface for the 'project' job after a build has been triggered. The 'Build Now' button is still present. The 'Stage View' now displays a progress bar for the 'Average stage times' with three stages: 'Declarative: Checkout SCM' (1s), 'Checkout' (1s), and 'Build Docker Image' (17s). Below this, a table shows the build progress for the current build (#1) on May 19, 2026, at 5:28 PM. The table has three columns: 'Declarative: Checkout SCM', 'Checkout', and 'Build Docker Image'. The first two columns show '1s' and the third shows '17s'. The 'Builds' section at the bottom left shows a list of builds, with the current build (#1) highlighted. The bottom of the browser window shows the Windows taskbar with the search bar and various application icons. The system tray shows the date and time as 10:59 PM on 5/19/2026.

To visualize the pipeline like above, install Blue ocean plugin from manag Jenkins & plugins installation page.

Also you can see the pipeline flow from console output to view the process

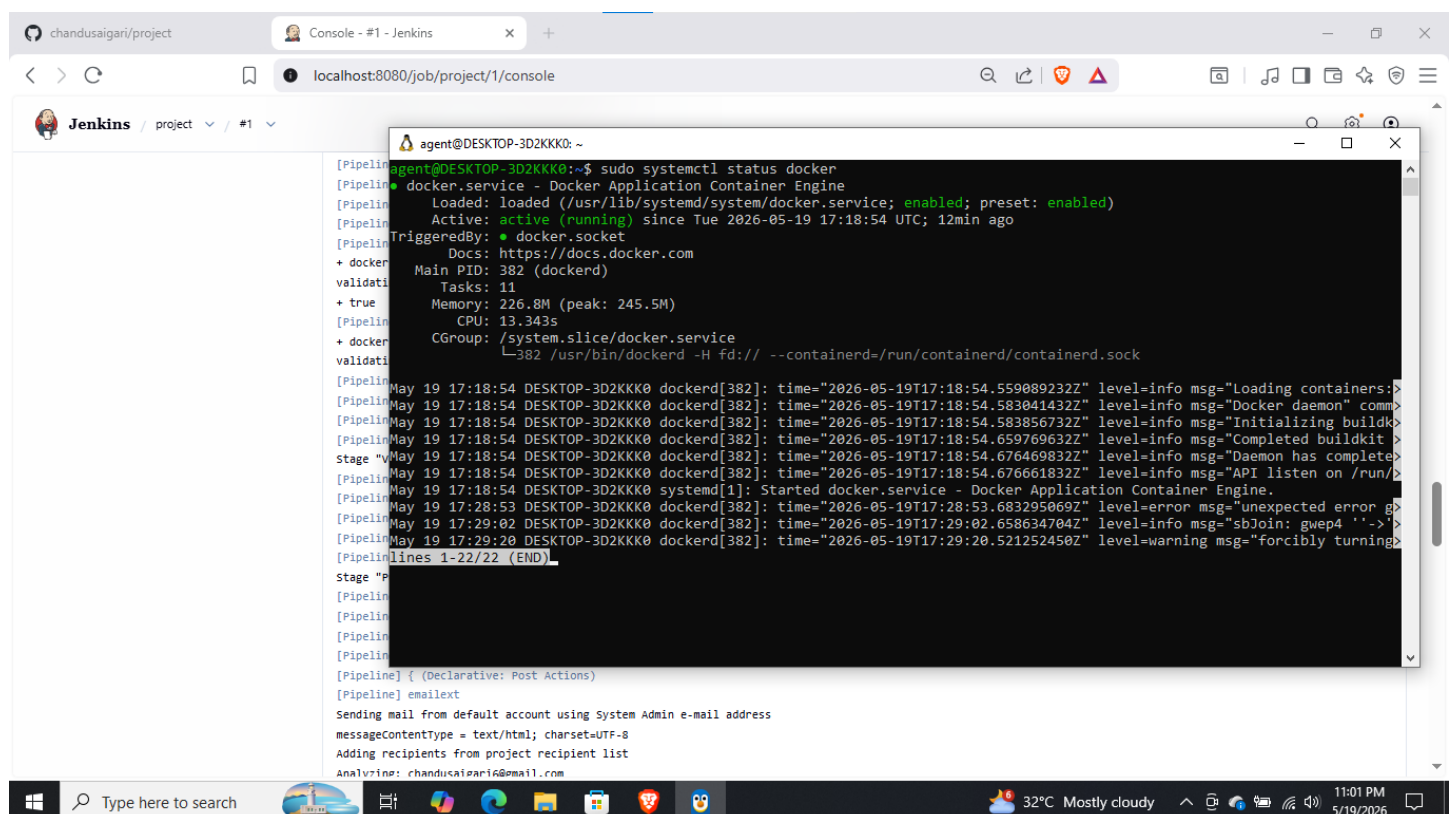
Benefits of viewing console is to know the failure at what point it is happened



The screenshot shows the Jenkins console output for a pipeline. The output includes the following steps and commands:

```
Started by user saigarichandu
Obtained Jenkinsfile from git https://github.com/chandusaigari/project
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/project
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected git installation does not exist. Using Default
The recommended git tool is: NONE
using credential git
> /usr/bin/git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/project/.git # timeout=10
Fetching changes from the remote Git repository
> /usr/bin/git config remote.origin.url https://github.com/chandusaigari/project # timeout=10
Fetching upstream changes from https://github.com/chandusaigari/project
> /usr/bin/git --version # timeout=10
> git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials
> /usr/bin/git rev-parse refs/remotes/origin/main^(commit) # timeout=10
Checking out Revision b7a4f048b65fb2c7d6eb1aa934d653baebf8a2 (refs/remotes/origin/main)
> /usr/bin/git config core.sparsecheckout # timeout=10
> /usr/bin/git checkout -f b7a4f048b65fb2c7d6eb1aa934d653baebf8a2 # timeout=10
Commit message: "Update docker-compose.yml"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
```

Make sure the docker is also running in background if not run the command in your terminal as shown below.



The screenshot shows the Jenkins console output with a terminal window overlay. The terminal window displays the command `sudo systemctl status docker` and its output:

```
agent@DESKTOP-3D2KKK0:~$ sudo systemctl status docker
docker.service - Docker Application Container Engine
Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-05-19 17:18:54 UTC; 12min ago
TriggeredBy: ● docker.socket
Docs: https://docs.docker.com
Main PID: 382 (dockerd)
Tasks: 11
Memory: 226.8M (peak: 245.5M)
CPU: 13.343s
CGroup: /system.slice/docker.service
└─382 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

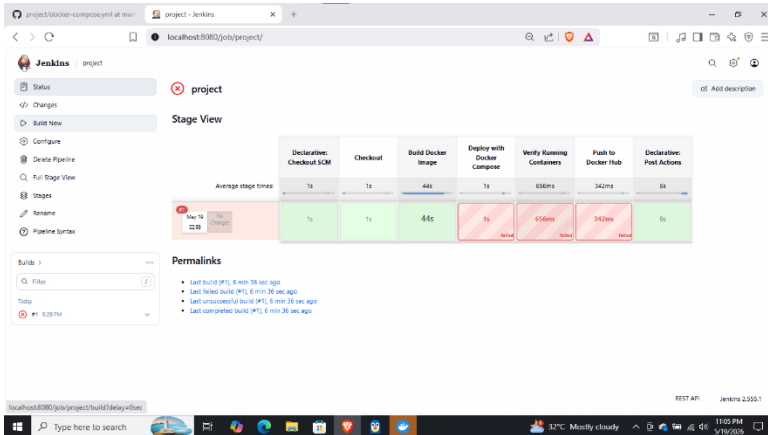
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.559089232Z" level=info msg="Loading containers: start."
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.583041432Z" level=info msg="Docker daemon" comm
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.583856732Z" level=info msg="Initializing buildkit"
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.659769632Z" level=info msg="Completed buildkit"
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.676469832Z" level=info msg="Daemon has complete
May 19 17:18:54 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:18:54.676661832Z" level=info msg="API listen on /run/2
May 19 17:18:54 DESKTOP-3D2KKK0 systemd[1]: Started docker.service - Docker Application Container Engine.
May 19 17:28:53 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:28:53.683295069Z" level=error msg="unexpected error g
May 19 17:29:02 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:29:02.658634704Z" level=info msg="sbJoin: gwep4 '->>
May 19 17:29:20 DESKTOP-3D2KKK0 dockerd[382]: time="2026-05-19T17:29:20.521252450Z" level=warning msg="forcibly turning
lines 1-22/22 (END)
```

Without docker running we cannot containerize the application of what we are building

After docker run, check the console for on going process

You can see we got a failure in pipeline

We need to check it in console output for knowing what is the actual error is



In console output

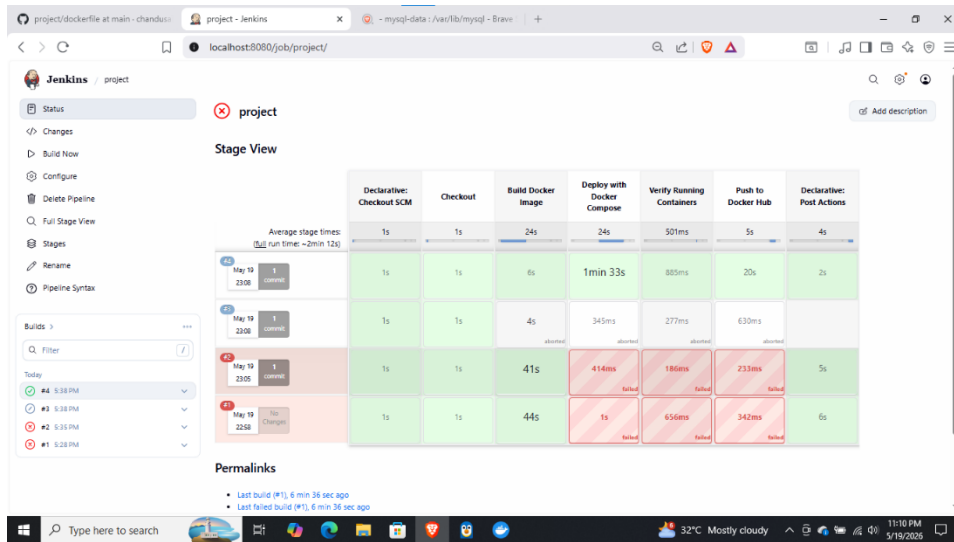
The screenshot shows the Jenkins Console View for the same pipeline. The console output is displayed in a scrollable area. The output shows the pipeline stages and the error message: 'validating /var/lib/jenkins/workspace/project/docker-compose.yml: volumes additional properties '/usr/local/bin/kind1' not allowed'. The pipeline is currently running on a node named 'k8s-1.19.2'. The console output for the 'Verify Running Containers' stage is as follows:

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy with Docker Compose)
[Pipeline] sh
+ docker compose down
validating /var/lib/jenkins/workspace/project/docker-compose.yml: volumes additional properties '/usr/local/bin/kind1' not allowed
+ true
[Pipeline] sh
+ docker compose up -d --build
validating /var/lib/jenkins/workspace/project/docker-compose.yml: volumes additional properties '/usr/local/bin/kind1' not allowed
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify Running Containers)
stage "Verify Running Containers" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Push to Docker Hub)
stage "Push to Docker Hub" skipped due to earlier failure(s)
[Pipeline] getContext
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] emailx
Sending mail from default account using System Admin e-mail address
messageContentType = text/html; charset=UTF-8
Adding recipients from project recipient list
AnalVriner: chandusaipari@gmail.com
```


It was saying that we did not allowed the Jenkins to access docker

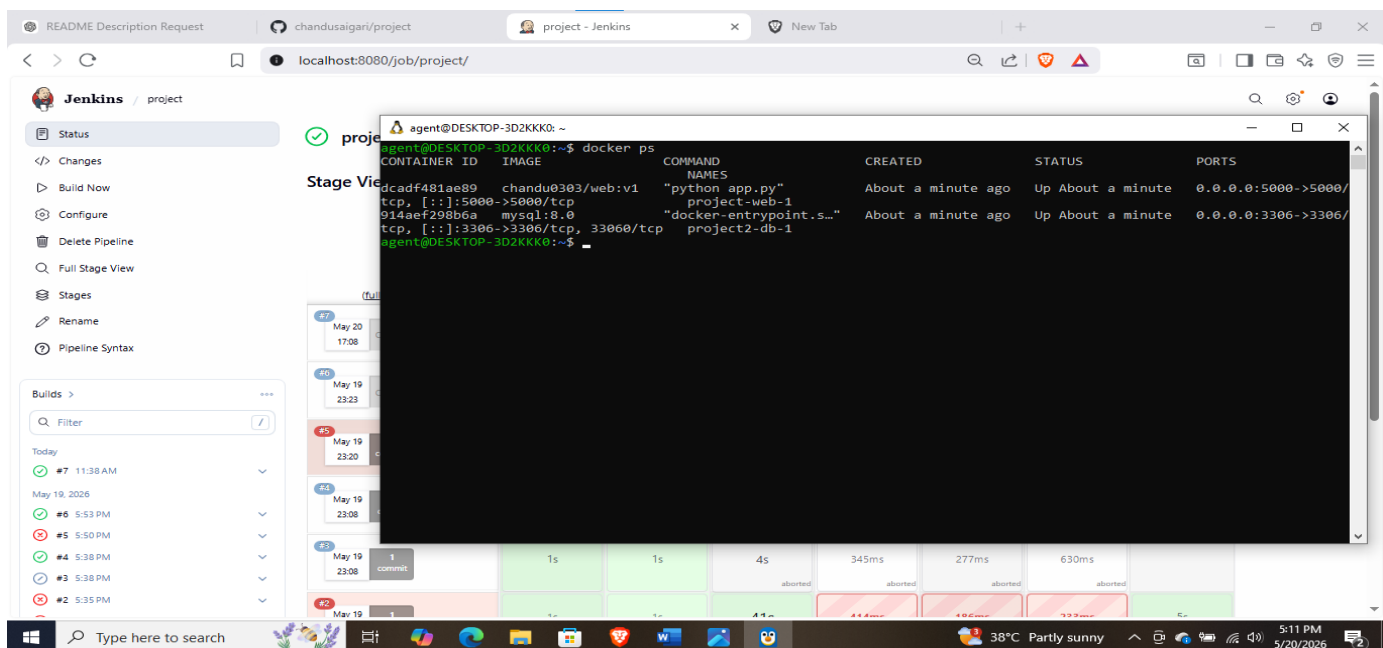
Solution is in terminal ,run the command :`sudo usermod -aG docker Jenkins`

By this we will give the access Jenkins from docker



As you can see our Job as successfully done.

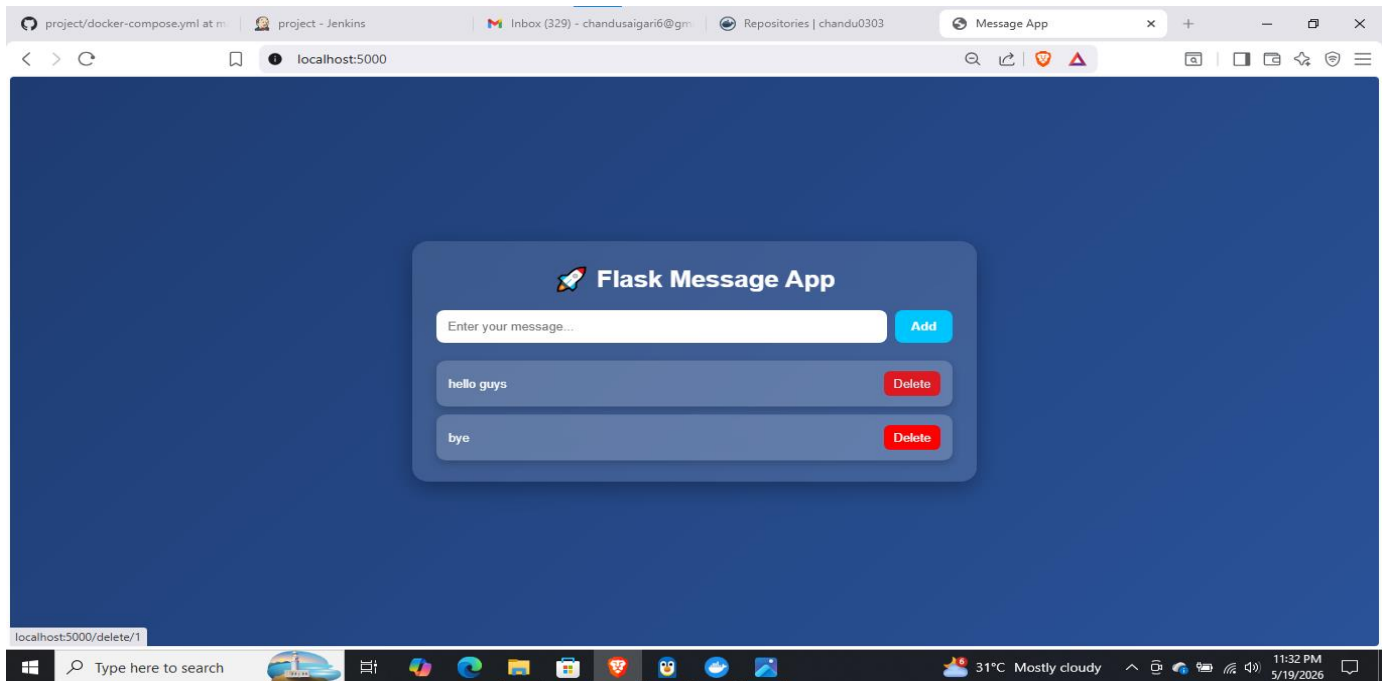
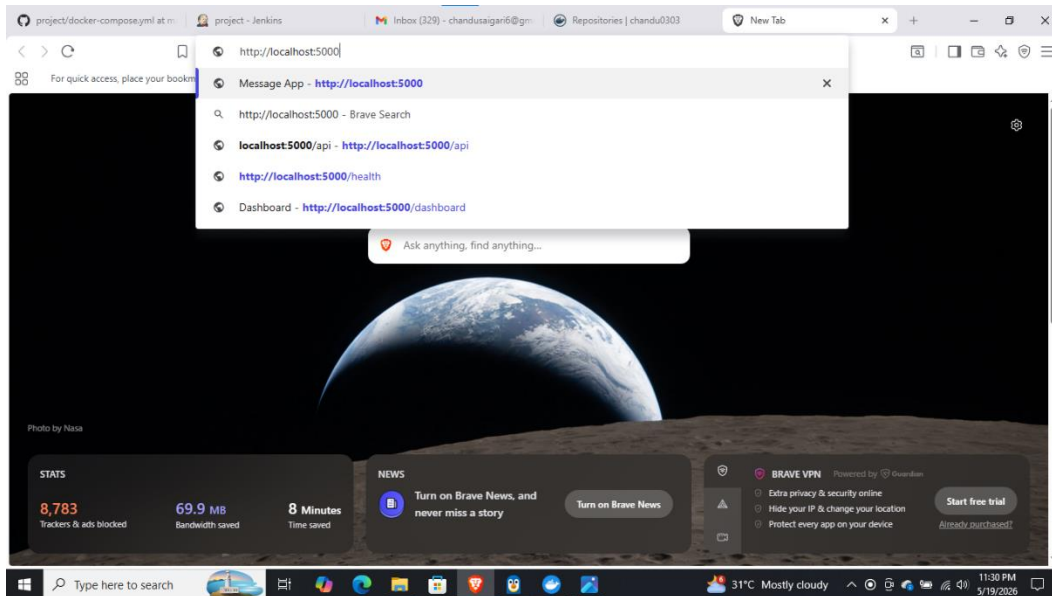
To check the Job done correctly or not we need to check the docker containers running and open your application from browser



Using `docker ps` command, you can check the running containers and also see at what port your application running

Mine is `localhost://5000`. Run this in web browser

you will get final running application like this



Check the YOUTUBE VIDEO for clear explanation:

Future Scope:

This project can be enhanced by migrating from Docker Compose to Kubernetes for better scalability, load balancing, auto-scaling, and self-healing capabilities. Kubernetes will help manage the application more efficiently in production environments.

For monitoring, Prometheus can be used to collect real-time metrics like CPU, memory, and request performance, while Grafana can visualize these metrics through dashboards for better system insights.

These improvements will make the application more scalable, reliable, and suitable for cloud-native production deployment.

Contact

**I am actively seeking opportunities to apply my skills in DevOps & related fields .
If you find my project interesting, I would be grateful for an opportunity to contribute and prove my abilities in a professional environment.**

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